PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Onesime Keya Kabiala
Program/Batch	B084-Networkwalks
Date	12 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

1. Liability Disclaimer

These activities were performed exclusively on systems I own or for which I have obtained explicit written authorization. This material is strictly for educational and research purposes. The instructor, authors, and Networkwalks assume no responsibility for any misuse of this knowledge. It is your sole responsibility to comply with the law. Unauthorized access is a criminal offense in most jurisdictions, even without resulting damage.

2. Introduction

This report details my Week 2 internship tasks at Networkwalks, focusing on two main phases: footprinting the networkwalks.com domain using Kali Linux (Module W2-PM1), and scanning my local network with Zenmap (Module W2-PM5). Together, these exercises demonstrate the transition from passive information gathering to active network mapping. Every step includes the exact command, observed results, screenshots, and an analysis of the findings from a security perspective.

3. Tools Used

Tool	Purpose
Kali Linux & Windows	Operating systems used for the practical exercises
WHOIS	Retrieve domain registration data (owner, dates, name servers)
WhatWeb	Identify web technologies (server, CMS, plugins)
nslookup	Resolve the target domain name to its IP address
curl -I	Inspect HTTP response headers
wafw00f	Detect the presence of a Web Application Firewall
dnsrecon	Enumerate DNS records (NS, MX, TXT, SRV, etc.)
Zenmap (Nmap GUI)	Scan the local network to discover live hosts and MAC addresses
Windows CMD	Identify local IP and MAC address

4. Activities Performed

4.1 Footprinting & Reconnaissance

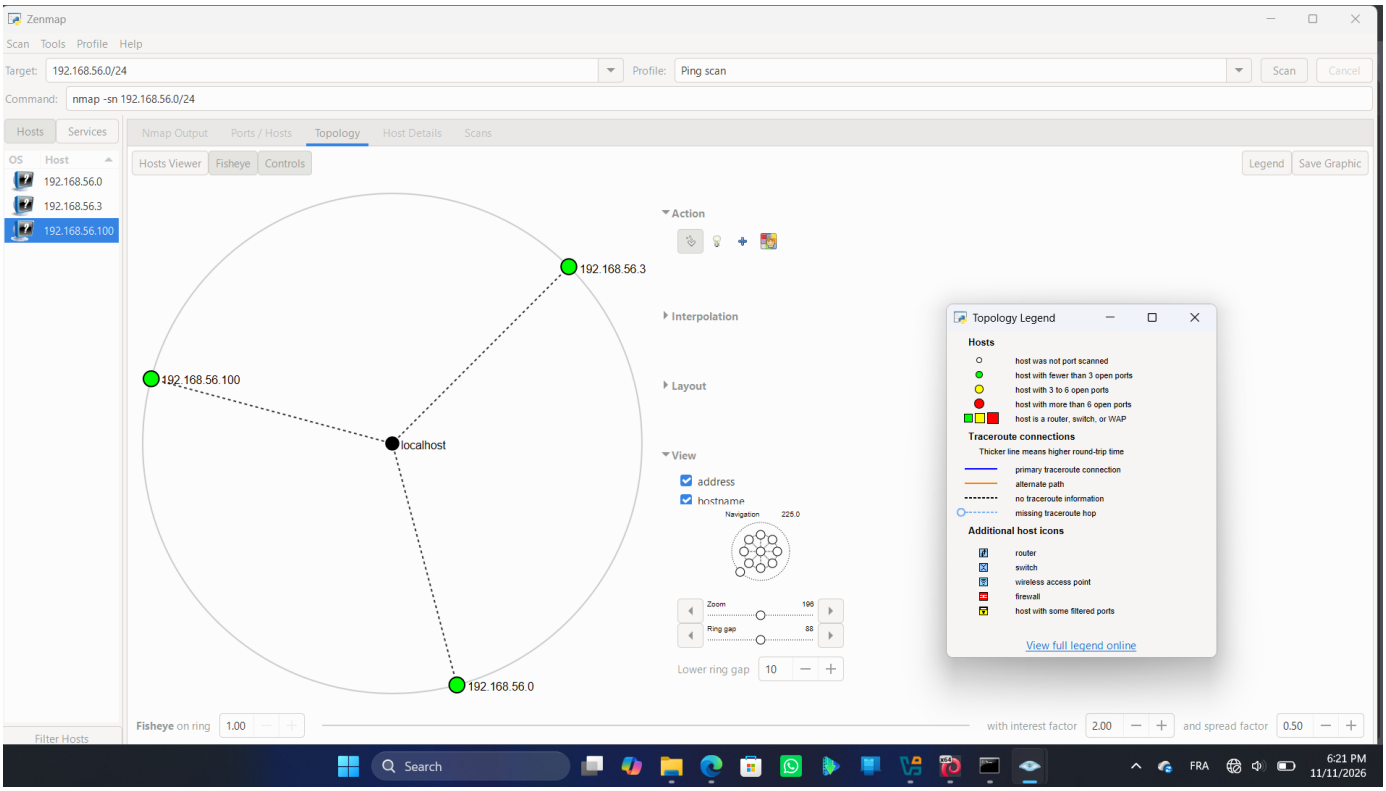
I used six Kali Linux tools to gather intelligence on networkwalks.com. First, **WHOIS** provided public domain registration details and name servers. Next, **WhatWeb** revealed the site uses **WordPress 7.1.4** and **WP Download Manager 3.3.76**.

I used **nslookup** to resolve the domain to IP **192.232.216.135**. Then, **curl -I** helped inspect HTTP headers, exposing the WordPress REST API endpoint (/wp-json/). **Wafw00f** detected a Web Application Firewall (**ModSecurity by SpiderLabs**). Finally, **dnsrecon** enumerated DNS records, yielding valuable data on mail servers and SPF/TXT records.

4.2 Network Scanning with Zenmap

For the second part, I scanned my local network using **Zenmap**. After finding my local IP and subnet via Windows ipconfig, I ran a **Ping Scan** in Zenmap.

The scan in my practical experience identified three active hosts (e.g., 192.168.56.100, 192.168.56.3, etc.) along with their MAC addresses. I then used Zenmap's **Topology** feature to generate and export a visual network map in PDF format, fulfilling the practical's requirements.



5. Risk Analysis / Impact

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Exposed web tech	WhatWeb found WordPress and plugins	Attackers can identify specific software versions to target	● **Medium**
2	Server IP exposed	nslookup resolved the domain IP	Reveals the network location of the web service	● **Low**
3	HTTP info exposed	Curl revealed response headers and /wp-json/	Aids in further technology fingerprinting	● **Low**
4	WAF identified	Wafw00f detected ModSecurity	Reveals the security architecture in place	● **Low**
5	DNS info exposed	DNSRecon found various DNS and mail records	Helps attackers map the broader infrastructure	● **Medium**
6	Live hosts visible	Zenmap found three active local devices	Unauthorized devices could be lurking on the network	● **Medium**

Risk level key: ● Critical ● Medium ● Low

Note: These are observational findings, not confirmed vulnerabilities. No exploitation was performed. Technical details alone do not imply a system is vulnerable without further authorized testing.

6. Recommendations

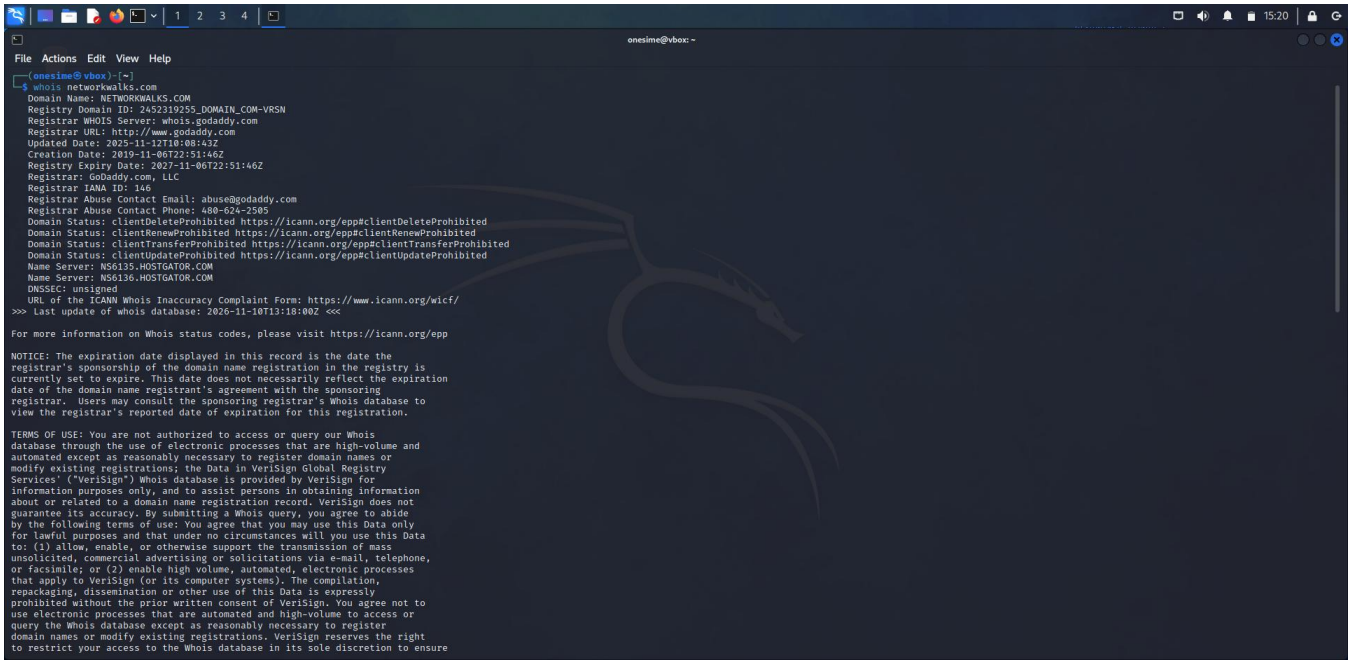
1. **Minimize public tech exposure:** Regularly audit what software and version info is visible publicly.
2. **Update software:** Keep CMS platforms and plugins patched against known advisories.
3. **Sanitize HTTP headers:** Ensure response headers don't leak unnecessary technical details.
4. **Review DNS records:** Periodically check public DNS records and remove unneeded ones.
5. **Maintain the WAF:** Keep ModSecurity enabled and properly tuned.
6. **Audit internal networks:** Perform regular scans to identify all active devices.
7. **Investigate anomalies:** Check any unexpected devices discovered during scans.
8. **Document the network:** Keep network topology and device inventories updated.
9. **Stay authorized:** Always ensure explicit permission is granted before scanning.

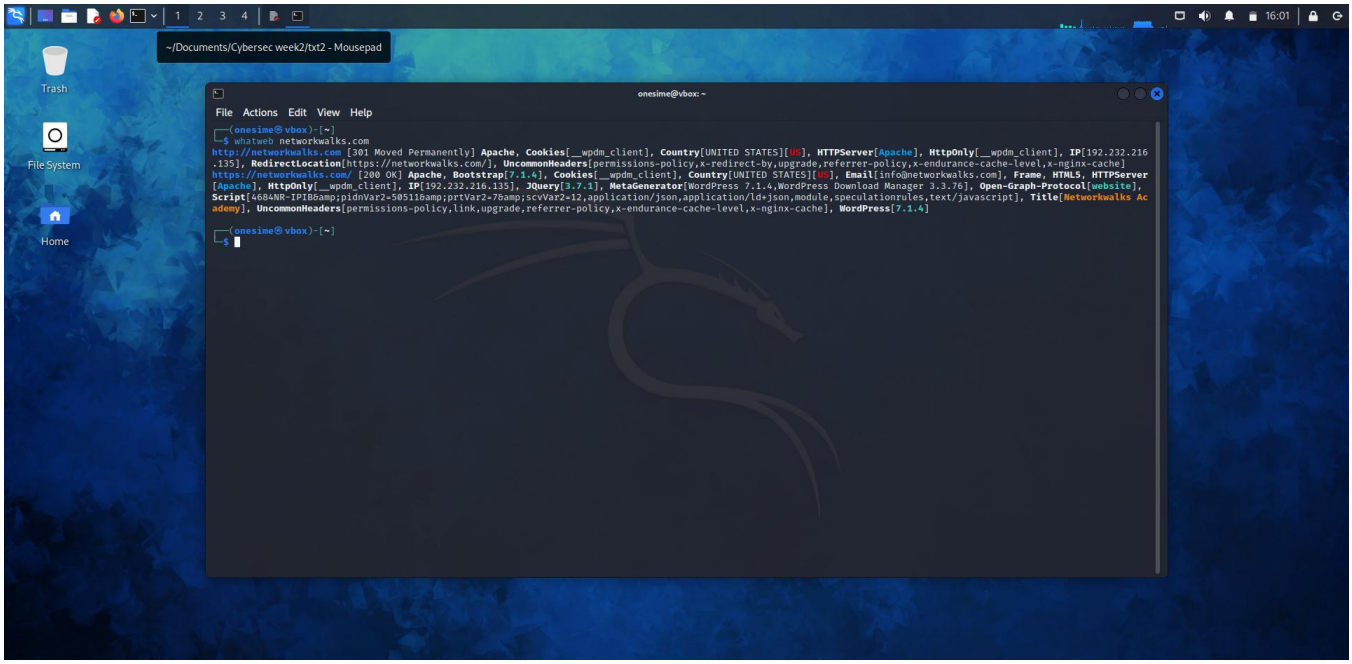
7. Conclusion

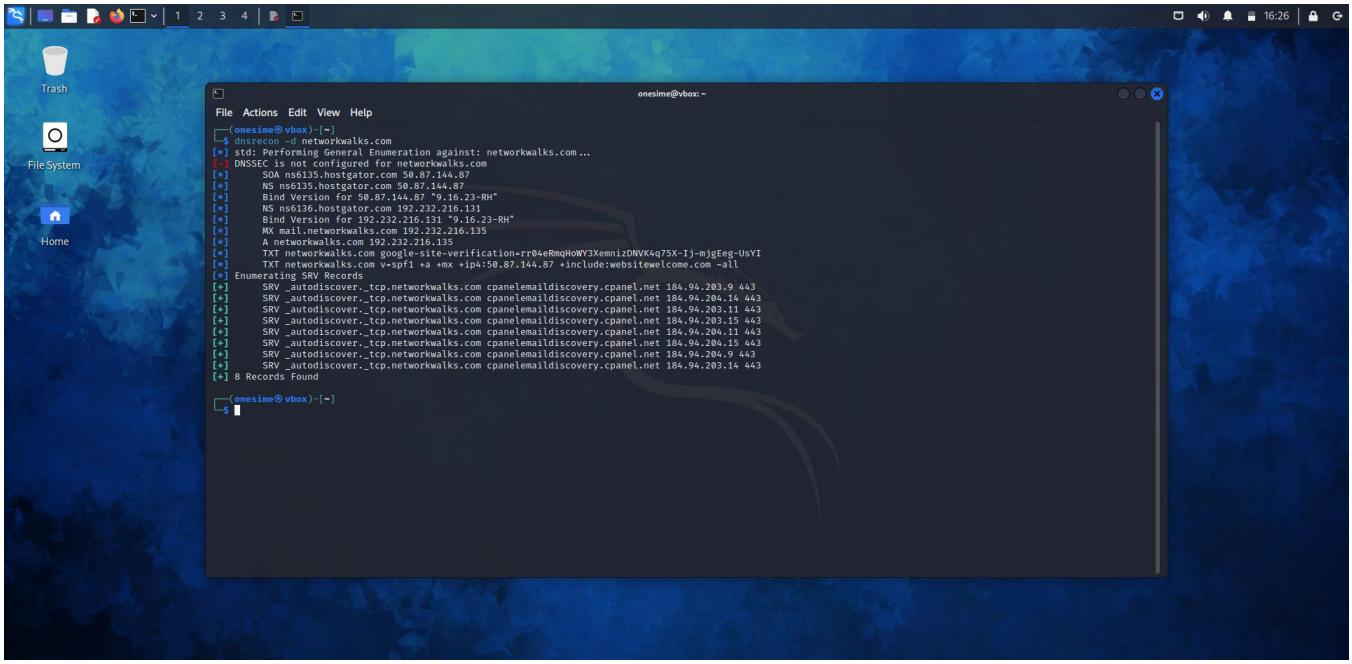
During Week 2, I completed practical exercises in footprinting and network scanning. Using Kali Linux tools, I learned to gather domain data, identify web technologies, resolve IPs, inspect headers, detect WAFs, and enumerate DNS. Using Zenmap, I successfully mapped active hosts on a local network and generated a topology map.

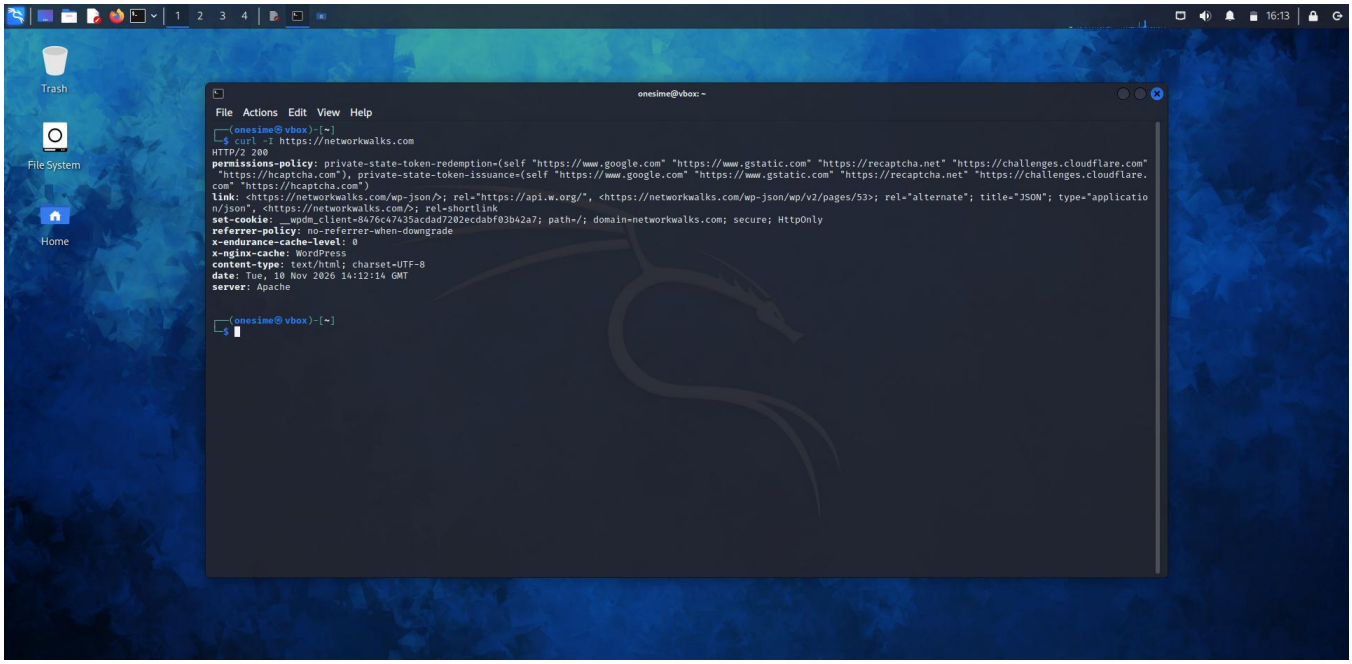
These exercises highlighted that information gathering is a foundational cybersecurity skill. I also learned the importance of documenting findings clearly, explaining what was done, what was found, the associated risks, and mitigation steps. Finally, I reinforced the absolute necessity of operating strictly within an authorized scope.

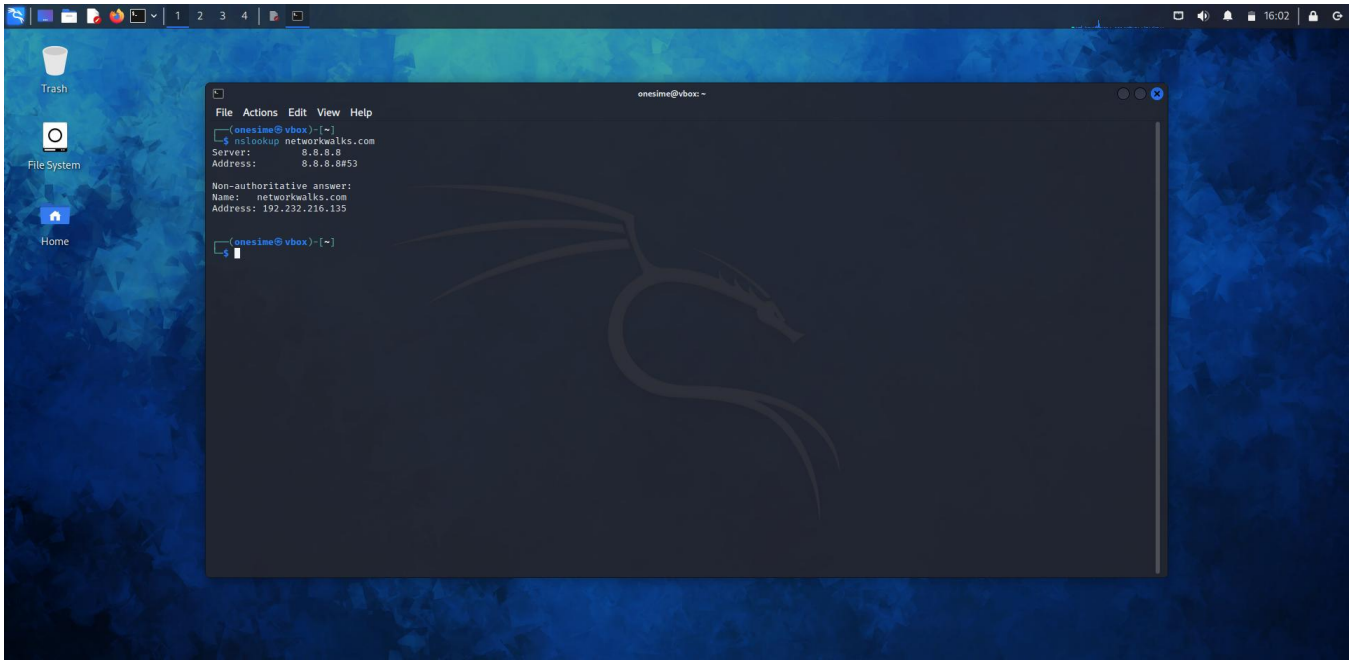
8. Evidences Collected

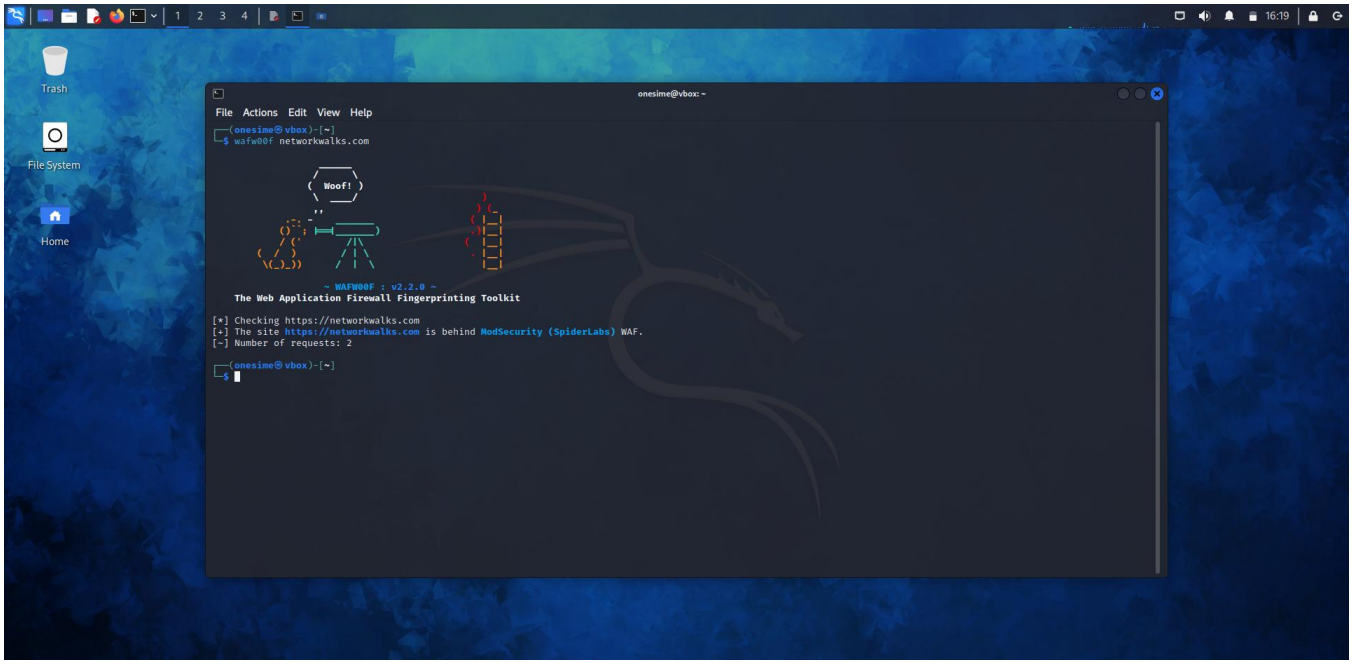


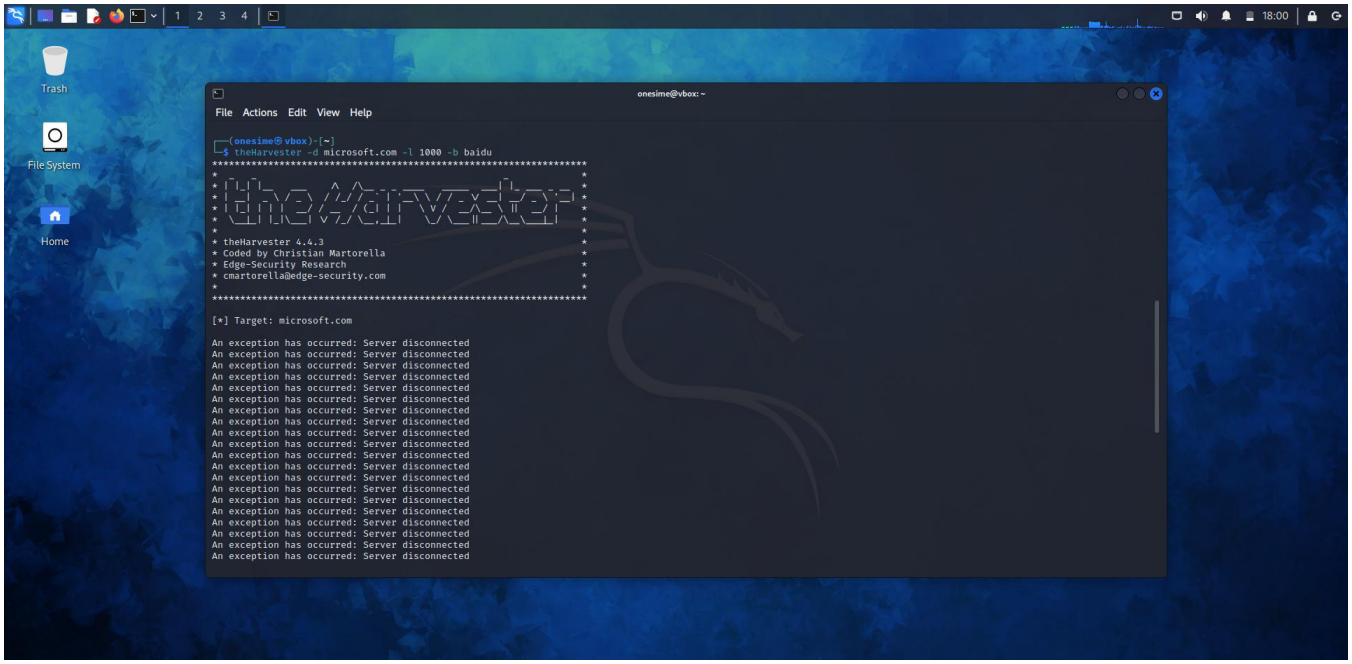


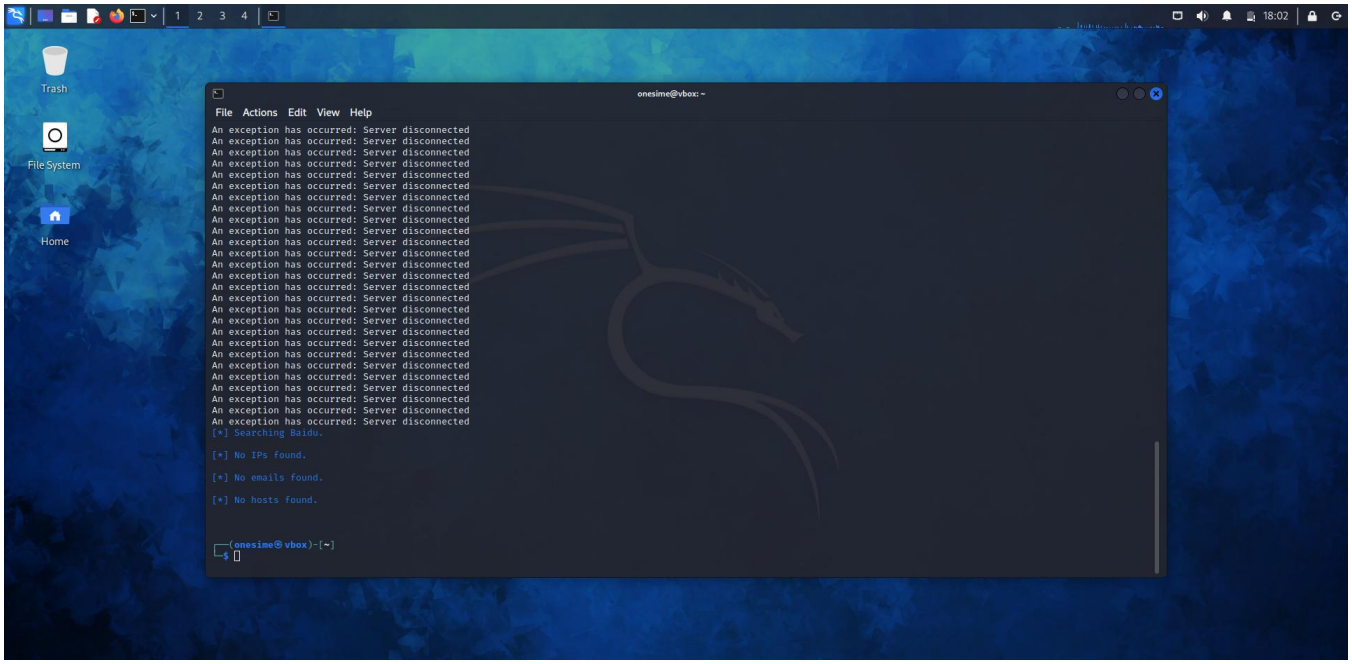

Trash
File System
Home
onesime@vbox: ~
File Actions Edit View Help
(onesime㉿vbox)-[~]
$ dnsrecon -d networkwalks.com
[*] std: Performing General Enumeration against: networkwalks.com ...
[-] DNSSEC is not configured for networkwalks.com
[*] SOA ns6135.hostgator.com 50.87.144.87
[*] NS ns6135.hostgator.com 50.87.144.87
[*] Bind Version for 50.87.144.87 "9.16.23-RH"
[*] NS ns6136.hostgator.com 192.232.216.131
[*] Bind Version for 192.232.216.131 "9.16.23-RH"
[*] MX mail.networkwalks.com 192.232.216.135
[*] A networkwalks.com 192.232.216.135
[*] TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
[*] TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
[*] Enumerating SRV Records
[+] SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.9 443
[+] SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.14 443
[+] SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.11 443
[+] SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.15 443
[+] SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.11 443
[+] SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.15 443
[+] SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.9 443
[+] SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.14 443
[+] 8 Records Found
(onesime㉿vbox)-[~]
$

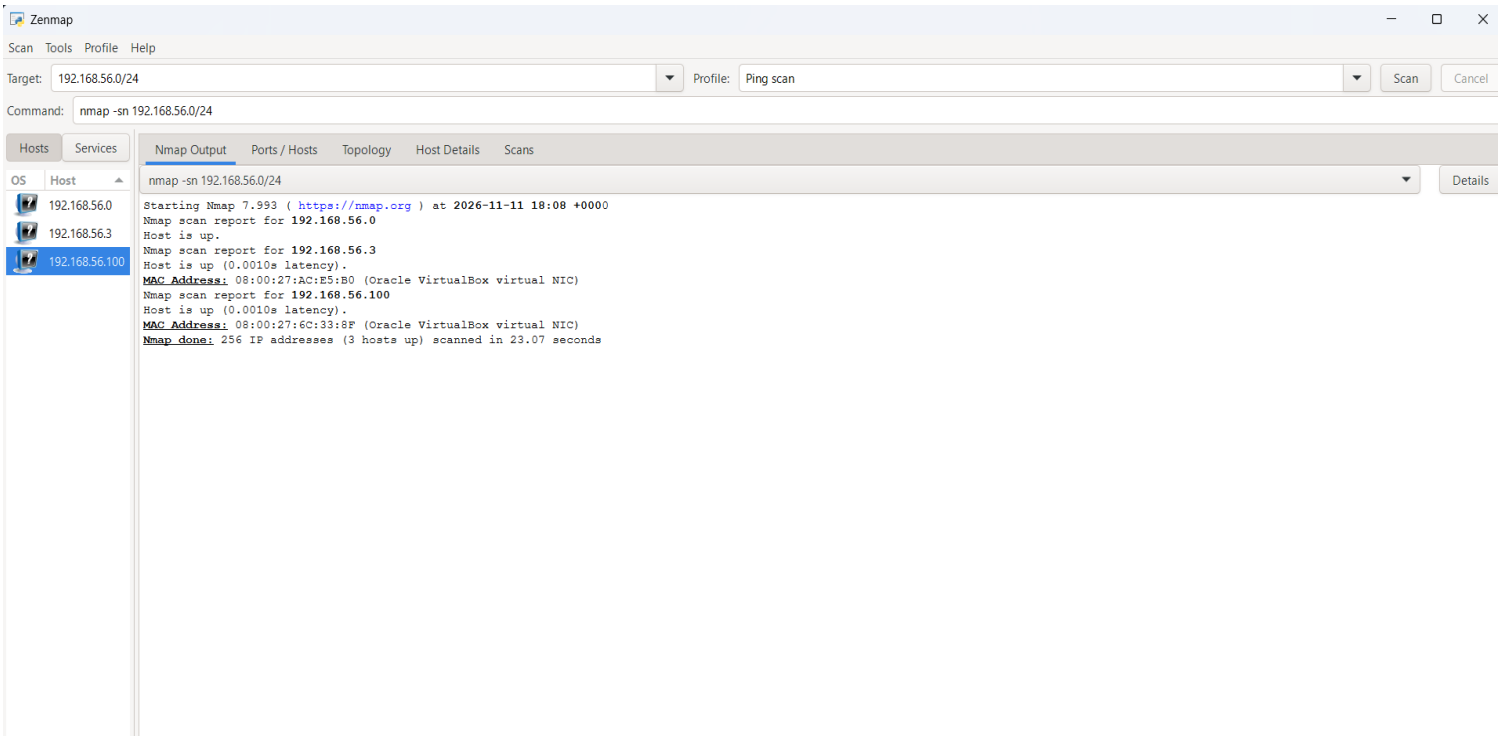

Trash
File System
Home
onesime@vbox: ~
File Actions Edit View Help
(onesime㉿vbox)-[~]
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=8476c47435acdad7202ecdabf03b42a7; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Tue, 10 Nov 2026 14:12:14 GMT
server: Apache
(onesime㉿vbox)-[~]
$

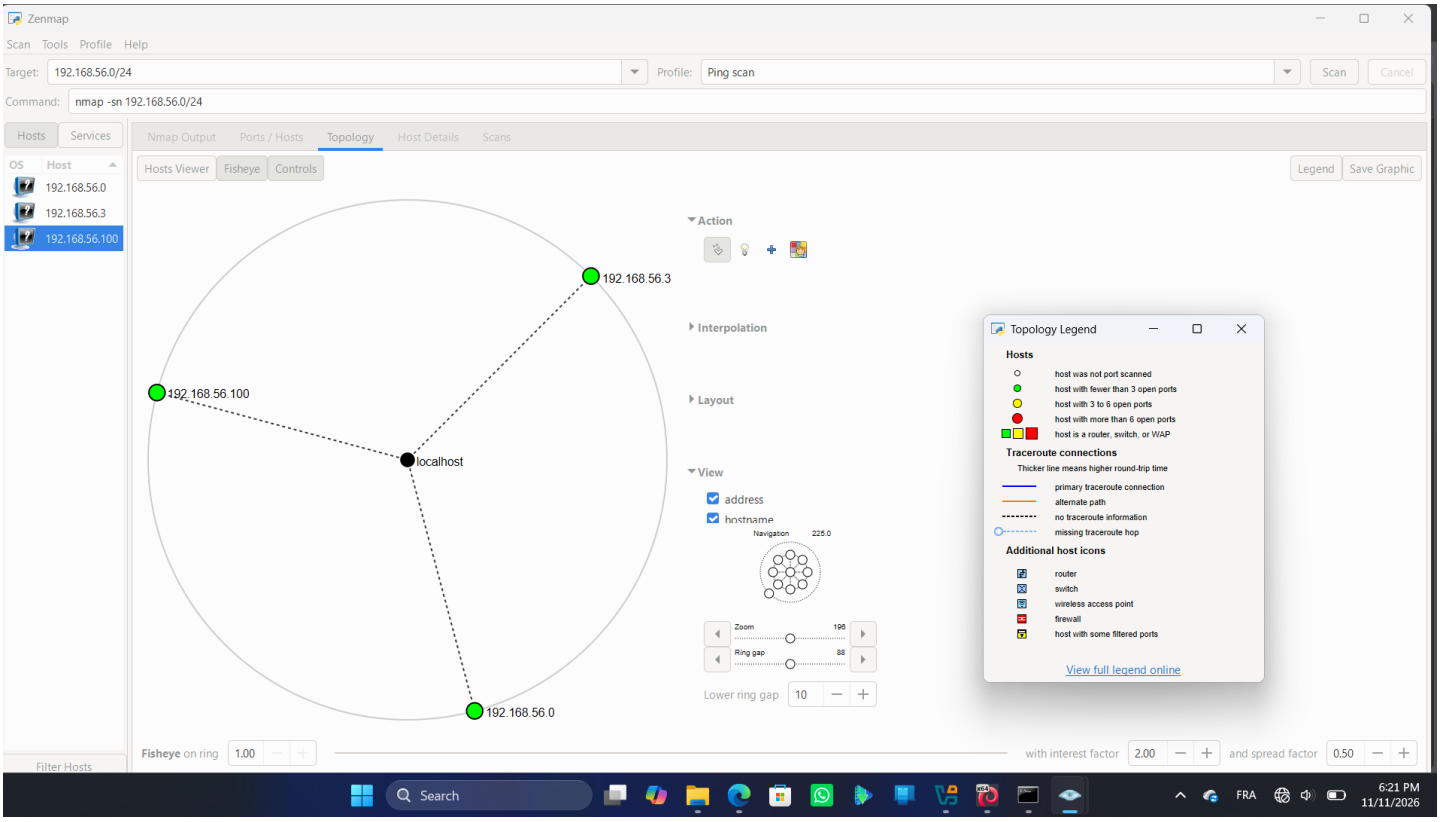

Trash
File System
Home
onesime@vbox: ~
File Actions Edit View Help
(onesime㉿vbox)-[~]
$ nslookup networkwalks.com
Server: 8.8.8.8
Address: 8.8.8.8#53
Non-authoritative answer:
Name: networkwalks.com
Address: 192.232.216.135
(onesime㉿vbox)-[~]
$


Trash
File System
Home
onesime@vbox: ~
File Actions Edit View Help
(onesime㉿vbox)-[~]
$ wafw00f networkwalks.com
Woof!
~ WAFW00F : v2.2.0 ~
The Web Application Firewall Fingerprinting Toolkit
[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
(onesime㉿vbox)-[~]
$


Trash
File System
Home
onesime@vbox: ~
File Actions Edit View Help
(onesime㉿vbox)-[~]
$ theHarvester -d microsoft.com -l 1000 -b baidu
* theHarvester 4.4.3
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
[*] Target: microsoft.com
An exception has occurred: Server disconnected


Trash
File System
Home
onesime@vbox: ~
File Actions Edit View Help
An exception has occurred: Server disconnected
[*] Searching Baidu.
[*] No IPs found.
[*] No emails found.
[*] No hosts found.
(onesime㉿vbox)-[~]
$


Zenmap
Scan Tools Profile Help
Target: 192.168.56.0/24
Profile: Ping scan
Scan
Cancel
Command: nmap -sn 192.168.56.0/24
Hosts
Services
OS
Host
192.168.56.0
192.168.56.3
192.168.56.100
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -sn 192.168.56.0/24
Details
Starting Nmap 7.93 (https://nmap.org) at 2026-11-11 18:08 +0000
Nmap scan report for 192.168.56.0
Host is up.
Nmap scan report for 192.168.56.3
Host is up (0.0010s latency).
MAC Address: 08:00:27:AC:E5:B0 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.100
Host is up (0.0010s latency).
MAC Address: 08:00:27:6C:33:8F (Oracle VirtualBox virtual NIC)
Nmap done: 256 IP addresses (3 hosts up) scanned in 23.07 seconds


Zenmap
Scan Tools Profile Help
Target: 192.168.56.0/24
Profile: Ping scan
Scan
Cancel
Command: nmap -sn 192.168.56.0/24
Hosts
Services
OS
Host
192.168.56.0
192.168.56.3
192.168.56.100
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
Hosts Viewer
Fisheye
Controls
Legend
Save Graphic
192.168.56.3
192.168.56.100
localhost
192.168.56.0
Action
Interpolation
Layout
View
address
hostname
Navigation
225.0
Zoom
196
Ring gap
88
Lower ring gap
10
Topology Legend
Hosts
host was not port scanned
host with fewer than 3 open ports
host with 3 to 6 open ports
host with more than 6 open ports
host is a router, switch, or WAP
Traceroute connections
Thicker line means higher round-trip time
primary traceroute connection
alternate path
no traceroute information
missing traceroute hop
Additional host icons
router
switch
wireless access point
firewall
host with some filtered ports
View full legend online
Fisheye on ring
1.00
with interest factor
2.00
and spread factor
0.50
Filter Hosts
Search
FRA
6:21 PM
11/11/2026

-End-

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Project Information
Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub